

Jawaharlal Nehru Govt Engineering College Sunder Nagar Mandi (H.P)
Course Name: Probability Statistical & Numerical Techniques Course Code: MAFS-311
Examination: 1st Periodical Branch - CSE (AI & ML)
Question Paper for 1st Periodical Examination (Sept, 2024)

Time: 2 Hours

Max Marks: 50

Note: Attempt All Questions

Course Outcomes: At The End Of Course The Student Will Be Able To:

- To understand the basic probability concepts.
- To have an in-depth knowledge of standard distribution & hypothesis testing which can describe real-life phenomena.
- To introduce numerical methods for solving interpolation problems and finding the roots of system of equations.
- To analyze the various methods related to numerical differentiation & integration along with the solution of ordinary differential equations.

Q.1 (CO1) (a) In a manufacturing plant, the probability of making a defective bolt is 0.1. Find the mean and standard deviation of defective bolts in a total of 900 bolts.

(b) Consider the continuous random variable with probability density function

$F(t) = 1+t$ for $-1 < t < 0$, $F(t) = 1-t$ for $0 < t < 1$. Compute the probability density function, Ex.

(c) Explain different types of sampling

(d) (CO2,3) Write short note on (1) Random variable (2) Bayes Theorem

(10)

Q.2 (CO1) The urn contains 7 white and 3 red balls, two balls are drawn together at random from the urn. Compute the probability that one of them is white; find the probability of getting one white and one red ball; hence compute expected number of white balls drawn.

(8)

Q.3 (CO1) The following data are number of seeds germinating out of 10 on damp filter paper for 80 sets of seeds. Fit a binomial distribution:

$X = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$, $f = 6, 20, 28, 12, 8, 6, 0, 0, 0, 0, 0$

(8)

Q.4 (CO2) A and B are two weak students of statistics and their chance of solving a problem in statistics correctly are $1/6$ and $1/8$ respectively. If the probability of their making a common error is $1/525$ and they obtain the same answer, find the probability that their answer is correct.

(8)

Q.5 (CO1) Let the joint pdf of x and y be $f(x,y) = (x+y)$, $0 \leq x \leq 1$, $0 \leq y \leq 1$. Find $E(X)$, $E(Y)$, $E(XY)$, $E(X+Y)$, $\text{var}(x)$, $\text{var}(y)$. Find $P(0 < x < 1/2 \mid 0 < y < 1/4)$.

(8)

Q.6 (CO2) If X is a poisson variate such that $P(x=2) = 9P(X=4) + 90P(X=6)$. Find Mean.

(8)